PRODUCT USER MANUAL

For Black Sea Waves Analysis and Forecasting Product

BLKSEA_ANALYSISFORECAST_WAV_007_003

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RECORD TABLE

Issue	Date	§	Description of Change	Author	Validated By
1.0	20.11.16	all	First version of PUM doc for Waves prod.	S. Ciliberti, A. Behrens, J. Staneva	C. Derval
1.1	21.01.19	all	General revision in the framework of Q2/2019 and addition of static files description	R. Lecci, A. Behrens, J. Staneva	C. Derval
1.2	03.04.20	all	Change of target delivery time from 00UTC to 12UTC New template and new timeseries temporal coverage	R. Lecci, A. Behrens, J. Staneva	C. Derval
2.0	10.09.20	all	New product name including new dataset	J. Staneva, A. Behrens, M. Ricker, R. Lecci	C. Derval
3.0	10.09.21	all	New product	R. Lecci, J. Staneva, M. Ricker, A. Behrens	C. Derval
3.1	29.11.22	all	Addition of data assimilation	R. Lecci, J. Staneva, M. Ricker, A. Behrens	Copernicus Product Manageme nt
3.2	30.11.23	all	Addition of LATEMAR maximum variables; new template.	R. Lecci, M. Ricker, J. Staneva, A. Behrens	Copernicus Product Manageme nt
4.0	26.11.24	all	Addition of a second forecast cycle and domain extension to include Azov Sea	R. Lecci, M. Ricker, J. Staneva, A. Behrens	

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GLOSSARY AND ABBREVIATIONS

BLK	Black Sea
CF	Climate Forecast (convention for NetCDF)
CLS	Collecte Localisation Satellites
CMAP	CPC Merged Analysis of Precipitation
CMEMS	Copernicus Marine Environment Monitoring Service
CTD	Conductivity Temperature Depth
DAC	Dynamic Atmospheric Correction
ECMWF	European Centre for Medium-Range Weather Forecasts
EOF	Empirical Orthogonal Function
FAQ	Frequently Asked Question
Hereon	Helmholtz-Zentrum Hereon (former HZG)
HZG	Helmholtz-Zentrum Geesthacht (outdated)
Meridional Velocity	South to North component of the horizontal velocity vector
MFC	Monitoring and Forecasting Centre
NEMO	Nucleus for European Modelling of the Ocean
NetCDF	Network Common Data Form
NOAA	National Oceanic and Atmospheric Administration
OA	Objective Analyses
OCEANVAR	Oceanographic variational data assimilation scheme developed at INGV/CMCC.
OGCM	Ocean General Circulation Model

OSI	Ocean and Sea Ice
PU	Production Unit
TAC	Thematic Assembly Centre
WAM	WAve Model
Zonal Velocity	West to East component of the horizontal velocity vector

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If you have any questions, please contact us.

I INTRODUCTION

I.1 Summary

This document is the user manual for the Copernicus Marine analysis and forecast product BLKSEA_ANALYSISFORECAST_WAV_007_003. It provides aggregated analyses updated twice per day with 10-day forecast. A 2-years rolling archive is available on the Marine Data Store.

An archive of forecasts is available, see Help Centre Article.

The wave products are the integrated parameters computed from the total wave spectrum (significant wave height, periods, directions, Stokes drift, etc), as well as the following partitions: the wind wave, the primary swell wave, and the secondary swell wave.

This product is defined on a standard grid at 1/40 degree (approx. 2.5km).

The product is organised in 2 datasets:

- cmems_mod_blk_wav_anfc_2.5km_PT1H-i containing 1-hourly instantaneous values for all the variables
- cmems_mod_blk_wav_anfc_2.5km_static containing the coordinates, mask, and bathymetry

It is updated as follows twice a day:

- the time series (analysis from the previous 12 hours and 10 days forecast) is updated at the end of the operational production process
- old obsolete forecasts are deleted from the server

The product is published on the MDS after automatic and human quality controls.

The analysis and forecasting system is described in the Quality Information Document (QUID).

Files downloaded are in NetCDF format and follow CF-1.6 convention.

I.2 History of changes

Date	Description of changes and impacted product
Jan 2019	adding the description of static files
Apr 2020	new template and new timeseries temporal coverage, change of target delivery time from 00UTC to 12UTC
Dec 2020	new product name and new dataset
Dec 2021	new product including new dataset
Nov 2022	product quality improvement (addition of data assimilation)
Nov 2023	addition of LATEMAR maximum variables
Nov 2024	Addition of a second forecast cycle and extension of the domain to include the Azov Sea

II DESCRIPTION OF THE PRODUCT SPECIFICATION

II.1 General Information

Product Line	BLKSEA_ANALYSISFORECAST_WAV_007_003
Geographical coverage	Black Sea: 27.25°E → 42.00°E; 40.50°N → 47.325°N
Product Type	Near Real Time - Analysis and Forecast
Variables	Spectral significant wave height (Hm0) Spectral moments (-1,0) wave period (Tm-10) Spectral moments (0,2) wave period (Tm02) Wave period at spectral peak / peak period (Tp) Mean wave direction from (Mdir) Wave principal direction at spectral peak Stokes drift U Stokes drift V Spectral significant wind wave height Spectral moments (0,1) wind wave period Mean wind wave direction from Spectral significant primary swell wave height Spectral moments (0,1) primary swell wave period Mean primary swell wave direction from Spectral significant secondary swell wave height Spectral moments (0,1) secondary swell wave period Mean secondary swell wave direction from

	Height of the highest crest Maximum crest trough wave height (Hc,max)
Available time series	-2Y to 10-days forecast
Temporal resolution	1-hourly instantaneous
Update Frequency	Twice-daily
Target delivery time	00:00/12:00 UTC
Horizontal resolution	1/40°
Number of vertical levels	Surface
Format	NetCDF4 CF1.6
Delivery mechanisms	MDS

II.2 Details of datasets

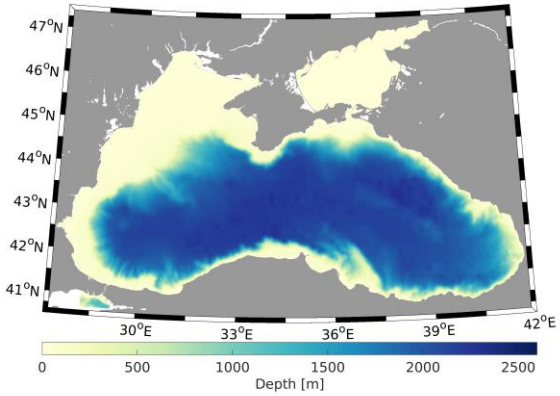
BLKSEA_ANALYSISFORECAST_WAV_007_003
Datasets: • **cmems_mod_blk_wav_anfc_2.5km_PT1H-i** contains all the variables • **cmems_mod_blk_wav_anfc_2.5km_static** contains the coordinates, mask, and bathymetry **Variables name in the NetCDF file and Unit: Long_name & Standard_name**
VHM0 [m] Spectral significant wave height (Hm0) sea_surface_wave_significant_height
VTM10 [s] Spectral moments (-1,0) wave period (Tm-10) sea_surface_wave_mean_period_from_variance_spectral_density_inverse_frequency_moment
VTM02 [s] Spectral moments (0,2) wave period (Tm02) sea_surface_wave_mean_period_from_variance_spectral_density_second_frequency_moment

VTPK [s] Wave period at spectral peak / peak period (Tp) sea_surface_wave_period_at_variance_spectral_density_maximum
VMDR [degree] Mean wave direction from (Mdir) sea_surface_wave_from_direction
VPED [degree] Wave principal direction at spectral peak sea_surface_wave_from_direction_at_variance_spectral_density_maximum
VSDX [m s-1] Stokes drift U sea_surface_wave_stokes_drift_x_velocity
VSDY [m s-1] Stokes drift V sea_surface_wave_stokes_drift_y_velocity
VHM0_WW [m] Spectral significant wind wave height sea_surface_wind_wave_significant_height
VTM01_WW [s] Spectral moments (0,1) wind wave period sea_surface_wind_wave_mean_period
VMDR_WW [degree] Mean wind wave direction from sea_surface_wind_wave_from_direction
VHM0_SW1 [m] Spectral significant primary swell wave height sea_surface_primary_swell_wave_significant_height
VTM01_SW1 [s] Spectral moments (0,1) primary swell wave period sea_surface_primary_swell_wave_mean_period
VMDR_SW1 [degree] Mean primary swell wave direction from

sea_surface_primary_swell_wave_from_direction
VHM0_SW2 [m] Spectral significant secondary swell wave height sea_surface_secondary_swell_wave_significant_height
VTM01_SW2 [s] Spectral moments (0,1) secondary swell wave period sea_surface_secondary_swell_wave_mean_period
VMDR_SW2 [degree] Mean secondary swell wave direction from sea_surface_secondary_swell_wave_from_direction
VMXL [m] Height of the highest crest sea_surface_wave_maximum_crest_height
VCMX [m] Maximum crest trough wave height (Hc,max) sea_surface_wave_maximum_height
e1t [m] Cell dimension along X axis
e2t [m] Cell dimension along Y axis
mask [1] Land-sea mask: 1 = sea ; 0 = land sea_binary_mask
depth0 [m] Bathymetry sea_floor_depth_below_geoid

Table 2: list of the datasets and variable names and unit for the BLKSEA_ANALYSISFORECAST_WAV_007_003 product

II.3 Production System Description

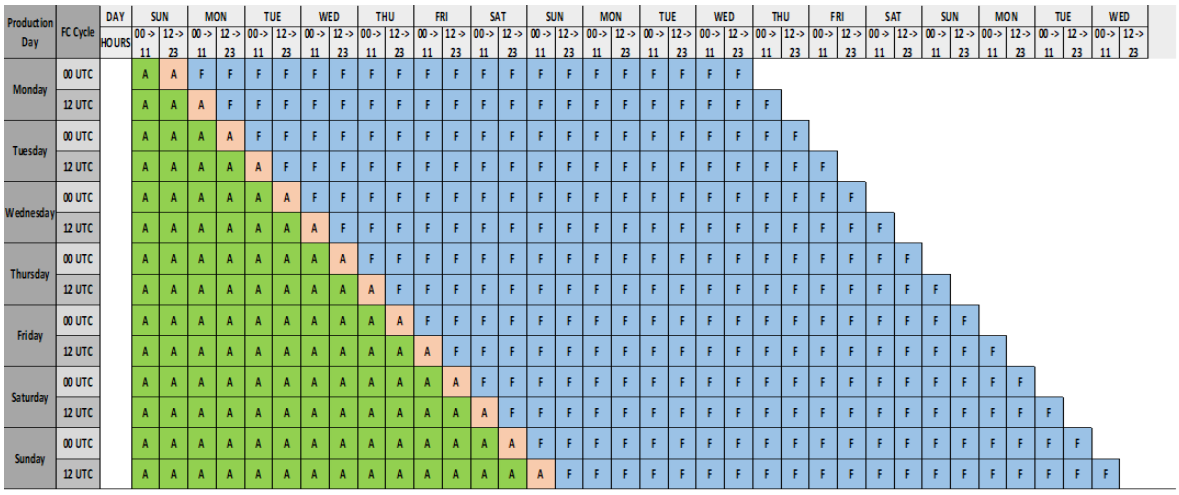
Domain **Resolution and grid** **Geographic coverage**	BLKSEA (27.25°E → 42.00°E; 40.50°N → 47.325°N) 1/40°; regular grid; 591 x 274 This product covers the Black Sea area; the horizontal resolution is approx. 2.5 km 
System Name	BSeas7
Model version **Production unit** **Period of validation**	• WAM Cycle 6 • Hereon, Germany • 01.05.2019-30.04.2021
Spectral resolution	24 directions, 30 frequencies
Interpolation method	Linear
Coupling	N/A
Bathymetry	GEBCO 30 sec interpolated on the model grid for the overall basin
Initial conditions	Initial 2-d-wave spectrum generated via fetch law from local wind field
Atmospheric forcing	ECMWF atmospheric forcing at 1/10 degree: 6-hourly analysis and 1-hourly for the first 3 days of forecast, 3-hourly for the next 3 days, and 6-hourly for the last 4 days.
Surface forcing	Sea surface height and surface currents from BLK-PHY (BLKSEA_ANALYSISFORECAST_PHY_007_001)

Boundary forcing	N/A
Data Assimilation scheme	Sequential optimal interpolation
Assimilated observations	Significant wave height and wind speed from altimetry data

The third-generation spectral wave model WAM Cycle 6 has been adapted to the Black Sea area and runs successfully on the cluster at Hereon. The shallow water version is implemented on a spherical grid with a spatial resolution of about 2.5 km (1/40 x 1/40°) with 24 directional and 30 frequency bins. The number of active wave model grid points is 81531. The model considers depth refraction and wave breaking and provides currently a 12-hourly analysis, followed by ten-days forecast with one-hourly output twice a day. The atmospheric forcing is taken from server dtn01.cmcc.it in NetCDF and is locally (on strand.hereon.de) transformed into a corresponding format required by the wave model WAM. Surface currents and sea surface height from BLKSEA_ANALYSISFORECAST_PHY_007_001 are used for additional forcing.

II.3.1.1 Production Cycle

BLKSEA_ANALYSISFORECAST_WAV_007_003 product temporal coverage: For everyday J, the available time series (hourly fields) start from 2 years back, up to the day J+10. The last ten days of the time series are forecast fields, while the remaining days of the archive are analysis. Every day, the wave system is run twice and each cycle provides 12 hours of analysis and ten days of forecast.



A **Aggregated time-series of hourly analysis** **A** **Analysis 1hr instantaneous**

F **Forecast 1hr instantaneous**

Figure 1: BLK-WAV production chain.

II.3.1.2 Observational Data

BLK-WAV provides 12-hourly analysis and 10-days forecasts updated twice per day. During the analysis phase, observations are assimilated into the system using the inherent data assimilation scheme of WAM Cycle 6.

Assimilated data: WAVE_GLO_PHY_SWH_L3_NRT_014_001 distributed by the CMEMS WAVE TAC.

II.4 Grid

BLK-WAV analysis and forecast product is delivered on a regular grid (coinciding with the native model grid) covering the whole Black Sea at 1/40° resolution. The grid covers the geographical area 27.25°E → 42.00°E; 40.50°N → 47.325°N and its size is 591 x 274. All wave variables are provided at the same horizontal resolution (1/40°, approx. 2.5km)

II.5 Processing information

II.5.1 Update Time

The product is updated twice a day at 00:00 UTC and 12:00 UTC.

II.5.2 Temporal extent of analysis and forecast stored on delivery mechanism

An archive of analysis over the last two years up to real-time is available.

II.5.3 Time averaging

The fields are 1-hourly instantaneous at 00, 01, 02, 03, …, 23 UTC.

III FILE FORMAT

The products are stored using the NetCDF format. To know more about the NetCDF format, please follow this link:

What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:

how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

IV FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this articles *" How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center"*

IV.1 Nomenclature of files when downloaded through the Copernicus Marine Web Portal subset Service

Products files nomenclature when downloaded through the CMEMS MDS Services is based on the following nomenclature:

The scheme is: **datasetname_nnnnnnnnnnnnn.nc**

where :

- **datasetname**: as described previously
- **nnnnnnnnnnnnn**: 13 digit integer corresponding to the current time (download time) in milliseconds since January 1, 1970 midnight UTC.
- **.nc**: standard NetCDF filename extension.

Example: cmems_mod_blk_wav_anfc_2.5km_PT1H-i_1303461772348.nc

IV.2 Nomenclature of original files The scheme is:

{valid date}{start time}_{freq flag}-{producer}--{parameter}-{config}-{region}-{bul date}_{product type}-sv{file version}.nc

Where

- **valid date** YYYYMMDD is the validity day of the data in the file
- **start time** HH is the hour (00 or 12 UTC) of the first date in the file
- **freq flag** is the frequency of data values in the file (h = hourly)
- **producer** is a short version of the Copernicus Marine production unit
- **parameter** is a four-letter code for the parameter or parameter set from Standard BODC
- **config** identifies the producing system and configuration
- **region** is a six-letter code for the region
- **bul date** bYYYYMMDD is the bulletin date the product was produced
- **product type** is a two-letter code for the product type, for example fc for forecast, an for analysis.
- **file version** is xx.yy where xx is the Copernicus Marine version and yy is an

incremental version number

Table 1 shows the nomenclature for the BLKSEA_ANALYSISFORECAST_WAV_007_003 product.

Table 1 Description of the nomenclature for BLKSEA_ANALYSISFORECAST_WAV_007_003

valid date	YYYYMMDD
start time	HH (00 for hourly fields beginning at 00:00 UTC of day DD; 12 for hourly fields beginning at 12:00 UTC of day DD)
freq flag	h (hourly)
producer	Hereon
config	BSeas7
region	BLK
parameter	WAVE
bul date	bYYYYMMDD
product type	an00 (analysis, produced by the 1st cycle) an12 (analysis, produced by the 2nd cycle) fc00 (forecast, produced by the 1st cycle) fc12 (forecast, produced by the 2nd cycle)
file version	10.00

Note that each forecast cycle produces 24 x 10 hourly forecast fields and that forecast files contain 12 timesteps each.

Examples for forecast files:

2022112900_h-Hereon--WAVE-BSeas7-BLK-b20221129_fc00-sv10.00.nc.

This file contains the 12 hourly instantaneous fields of the wave parameters produced by forecast cycle 1 (fc00), from 00:00 UTC of the 29th of November 2022 to 11:00 UTC of the 29th of November 2022.

2022112912_h-Hereon--WAVE-BSeas7-BLK-b20221129_fc12-sv10.00.nc.

This file contains the 12 hourly instantaneous fields of the wave parameters produced by forecast cycle 2 (fc12), from 12:00 UTC of the 29th of November 2022 to 23:00 UTC of the 29th of November 2022.

Note that two analysis files per day are available, containing 12 timesteps each.

Examples for analysis files:

2022112900_h-Hereon--WAVE-BSeas7-BLK-b20221129_an12-sv10.00.nc.

This file contains the 12 hourly instantaneous fields of the wave analysis (an12) from 00:00 UTC of the 29th of November 2022 to 11:00 UTC of the 29th of November 2022.

2022112912_h-Hereon--WAVE-BSeas7-BLK-b20221130_an00-sv10.00.nc.

This file contains the 12 hourly instantaneous fields of the wave analysis (an00) from 12:00 UTC of the 29th of November 2022 to 23:00 UTC of the 29th of November 2022.

IV.3 Other information: mean centre of Products, land mask value, missing value

Real_Value = (Display_Value X scale_factor) + add_offset

The missing value for this product is: 1e+20

Land and sea-ice masks are equal to "_FillValue" (see variable attribute on NetCDF file).

In the Copernicus Marine Service online system data from the latest 2 years (running window) are available via these download interfaces: subset and FTP download.

IV.4 File size

DATASET NAME	FILE NAME	DIMENSION [MB]

cmems_mod_blk_wav_anfc_2.5km_PT1H-i	{date1}{00|12}_h-Hereon--WAVE–BSeas7-BLK-b{date2}_an{00|12}-sv10.00.nc {date1}{00|12}_h-Hereon--WAVE-BSeas7-BLK-b{date2}_fc{00|12}-sv10.00.nc	47
cmems_mod_blk_wav_anfc_2.5km_static	BLK-MFC_007_003_${field}.nc	2

V REFERENCES

QUID – Quality Information Document :
https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-BLK-QUID-007_003.pdf

Citation of the dataset:

Staneva, J., Ricker, M., & Behrens, A. (2022). Black Sea Waves Analysis and Forecast (CMEMS BS-Waves, EAS5 system) (Version 1) [Data set]. Copernicus Monitoring Environment Marine Service (CMEMS). https://doi.org/10.25423/CMCC/BLKSEA_ANALYSISFORECAST_WAV_007_003_EAS5